T7

Iowa State University ILL

ILLiad TN: 561774

Borrower: WPG

Lending String: FYC,MDN,*IWA,SMT,PMC

Patron:

Journal Title: Rare earths : extraction, preparation and applications : proceedings of a symposium on rare earths, extraction, preparation and applications /

Volume: **Issue:**
Month/Year: 1988 **Pages:** 355-368

Article Author: Sharma & Seefurth

Article Title: A molten salt process for producing neodymium-iron alloys by reduction of Nd2O3 with sodium

Imprint: Warrendale, Pa. : TMS, ©1988.

ILL Number: 217776839

Call #: TN799.R37 R37 1988
Location: General Collection

Charge
Maxcost: 25.00IFM

ODYSSEY ENABLED

Shipping Address:
Gordon Library ILL
WPG-Worcester Polytechnic Institute
100 Institute Road
Worcester, Massachusetts 01609
United States

Email: lib-ill@wpi.edu

NOTICE ON COPYRIGHT RESTRICTIONS

Title 17, U.S. Code, governs reproductions of copyrighted material. This institution reserves the right to refuse an order if, in its judgment, fulfillment would involve violation of copyright law.

A MOLTEN SALT PROCESS FOR PRODUCING NEODYMIUM-IRON ALLOYS BY REDUCTION OF Nd_2O_3 WITH SODIUM

R. A. Sharma and R. N. Seefurth

Physical Chemistry
General Motors Research Laboratories
Warren, Michigan 48090

Abstract

Neodymium metal was successfully produced by the reduction of Nd_2O_3 with Na in the presence of $CaCl_2$-NaCl melts by the overall reduction reaction $Nd_2O_3 + 3\ CaCl_2 + 6\ Na \rightarrow 2\ Nd + 3\ CaO + 6\ NaCl$ at ~750°C. The metal produced was extracted from the salt media by dissolving it in a Nd-Fe alloy pool. The yield (percentage of theoretical neodymium produced) was found to depend on the salt composition and the mass ratio of salt to neodymium in the reactor. A yield of 95% was obtained with salt ratios as low as 3.5. Further decrease in the salt ratio below 3.25 decreased the yield. Stirrer position had little effect on yield; while increasing the temperature and placing fins in the reactor increased the yield. The Nd-Fe alloy produced was of as good quality as that produced using Ca reductant, and was suitable for direct use in producing General Motors MAGNEQUENCH™ alloy for permanent magnets.

Rare Earths, Extraction,
Preparation and Applications
Edited by R.G. Bautista and M.M. Wong
The Minerals, Metals & Materials Society, 1988

Introduction

A process known as MAGNEQUENCH™* [1] has been developed at GM to produce powerful permanent magnets. The magnetic alloy used in this process consists of neodymium, iron and boron. A new process was proposed to produce neodymium metal at potentially lower cost than existing processes [2]. In this process, neodymium oxide (Nd_2O_3) was reduced with calcium in a calcium chloride-sodium chloride ($CaCl_2$-NaCl) molten salt media at ~750°C. The metal produced was extracted using a molten neodymium-zinc (Nd-Zn) or neodymium-iron (Nd-Fe) alloy pool of approximately eutectic composition and the results were communicated in previous publications [3,4].

Additional work was carried out using a Nd-Zn extraction pool and sodium instead of calcium as the reductant. This was done to make the production process more cost effective as sodium is one fifth as costly as calcium per pound. Results of these experiments have also been communicated [5,6]. However, in this case, zinc has to be removed from the Nd-Zn alloy before it can be used in the preparation of the MAGNEQUENCH™ alloy [1]. Therefore, in another set of experiments, the neodymium produced was extracted using a molten Nd-Fe alloy pool of nearly eutectic composition. Work was carried out to determine the effects of salt composition and amount on yield. These results are communicated here.

Theoretical Background

On the basis of the standard free energies of formation of sodium oxide (Na_2O) and Nd_2O_3 [7,8], neodymium metal can not be produced by the reduction of Nd_2O_3 with sodium by the following reaction,

$$Nd_2O_3 + 6\ Na \longrightarrow 2\ Nd + 3\ Na_2O \qquad (1)$$

However, on the basis of the standard free energies of formation of calcium oxide (CaO) and Nd_2O_3 presented in Fig. 1, neodymium metal can be produced by the following reaction,

$$Nd_2O_3 + 3\ Ca \longrightarrow 2\ Nd + 3\ CaO \qquad (2)$$

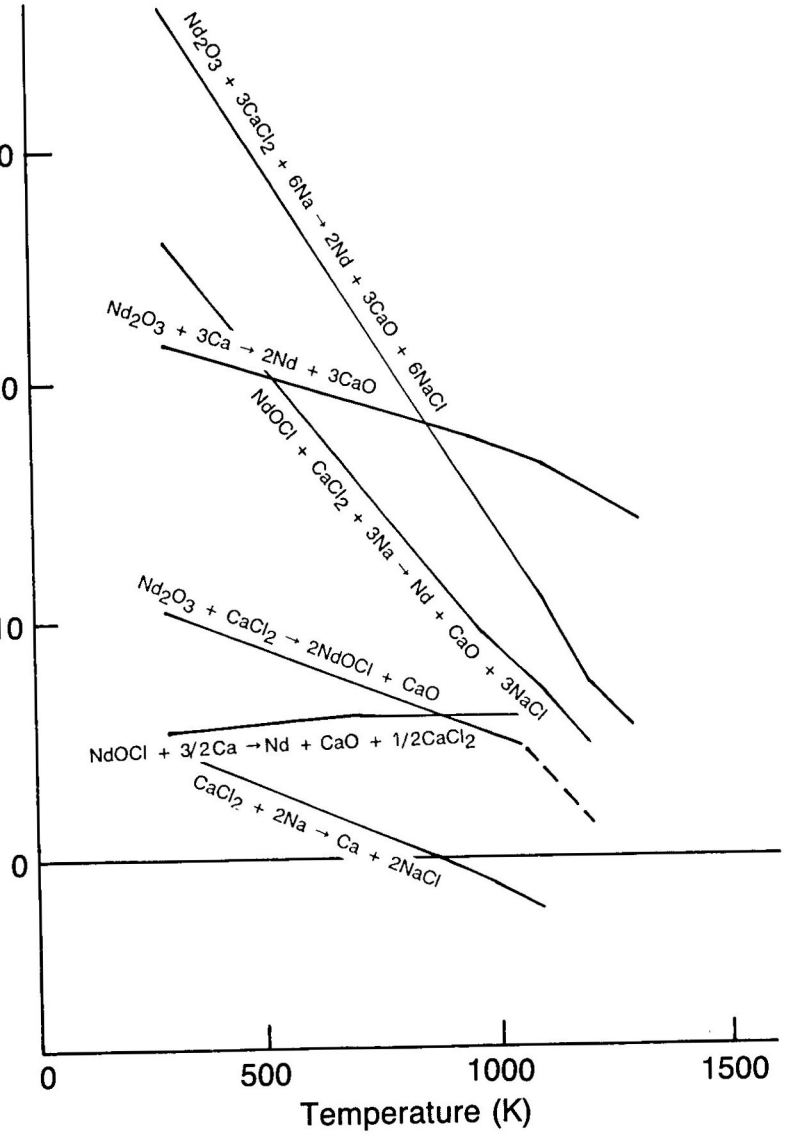
This has been confirmed experimentally [3,4].

Calcium required for reaction (2) may be generated by the reaction of $CaCl_2$ with sodium by the following reaction,

$$CaCl_2 + 2\ Na \rightleftharpoons Ca + 2\ NaCl \qquad (3)$$

---

*MAGNEQUENCH is a tradename of General Motors. The product is a hard magnetic material of general formula $Nd_{0.14}Fe_{0.78}B_{0.08}$.



1 Standard free energy changes, ΔG°, of the reactions as a function of temperature [7,8].

The standard free energy change of this reaction is negative below 900 K. Above this temperature, reaction (3), because of the positive value of the standard fee energy change, becomes doubtful.

The data presented in Ref. 2 indicate that melts of calcium and sodium coexist with melts of $CaCl_2$ and NaCl at various temperatures For example, a melt of 86 weight percent (w/o) Ca and 14 w/o Na coexists with a melt of 70 w/o $CaCl_2$ and 30 w/o NaCl at 710°C. This means a Ca-Na melt of 86 (w/o) Ca or greater can be maintained in a reactor by having a $CaCl_2$-NaCl melt of 70 w/o $CaCl_2$ or greater. Each of these Ca-Na melts has a Ca-activity high enough for carrying out the Nd_2O_3 reduction. The amounts of $CaCl_2$ required to have Ca-Na melts of 86 w/o or greater in the reactor during the reduction can be predetermined. Considering all the above information, it should be possible to produce neodymium metal by the following reaction,

$$Nd_2O_3 + 3\ CaCl_2 + 6\ Na \longrightarrow 2\ Nd + 3\ CaO + 6\ NaCl \qquad (4)$$

The standard free energy change of this reaction which is a combination of reactions (2) and (3) is negative at all temperatures of interest (Fig. 1).

However, it is experimentally found that Nd_2O_3 reacts with $CaCl_2$ in the salt phase to form neodymium oxychloride (NdOCl) by the following reaction,

$$Nd_2O_3 + CaCl_2 \longrightarrow 2\ NdOCl + CaO \qquad (5)$$

The standard free energy change of this reaction is negative as shown in Fig. 1, and so the reaction should proceed. Therefore, the actual reaction to produce neodymium becomes,

$$2\ NdOCl + 2\ CaCl_2 + 6\ Na \longrightarrow 2\ CaO + 6\ NaCl + 2\ Nd \qquad (6)$$

The standard free energy change of this reaction is also negative as shown in Fig. 1. So this reaction should proceed and neodymium should be produced.

Experimental

Materials

Neodymium-iron alloy pieces of eutectic [9] composition (99% purity) and iron metal powder (100 mesh) of 99% purity were used to make the extraction pool.

Sodium chloride of 99+% purity, potassium chloride (KCl) of 99.7% purity, and anhydrous $CaCl_2$ of 99.9+% purity were used to prepare the various composition salt baths. Prior to use, each salt was oven baked for ~2 h at 400°C to remove moisture.

Sodium metal of 99.9+% purity, and Nd_2O_3 powder of at least 95% purity (up to 5% praseodymium oxide) were used. Before reduction, the oxide was oven dried for ~2 h at 900°C to remove volatiles.

Apparatus

All experiments were carried out in a helium-atmosphere drybox using apparatus similar to that previously described [3,4] and shown in Fig. 2. Minor differences included the use of mild steel reaction vessels (10.2 cm I.D. x 17.8 cm deep x 0.15 cm wall) which had four fins spotwelded to the inner wall of the vessel 90° apart to control vortex formation during stirring.

Procedures

The iron-rich molten alloy pool to recover reduced neodymium metal from the salt phase was prepared by fusing the Nd-Fe eutectic pieces and iron powder together in the preweighed mild steel reaction vessel. A predetermined weight of the eutectic alloy was placed in the reaction vessel and the entire reaction assembly was lowered into the furnace well which was heated to 750°C. After the eutectic alloy was molten, the predetermined amount of iron powder was slowly mixed into the molten pool. At this time, the preweighed salt components were added into the reaction vessel and allowed to fuse.

Upon attaining operating temperature, a preweighed amount of Nd_2O_3 was added to the fluid melt. The stirrer was introduced, and used to wet and mix the oxide into the melt. A predetermined weight of the sodium reductant was inserted into the melt and the baffle-splash guard assembly was positioned. The stirrer bracket was slipped into place and the stirrer motor assembly adjusted into position. The stirrer blade was positioned to the prescribed height and secured in the flexible shaft chuck. The reaction melt was fast-stirred for an assigned time interval and then slow-stirred for an additional period of time. Experimentally, these were found necessary for good yields.

When the allotted time intervals passed, the stirrer motor was shut off, the stirrer blade carefully removed from the melt, and the reaction vessel pulled from the furnace well and quenched. The cold reaction vessel was sealed in double plastic bags, removed from the drybox and radiographs were taken. Complete reaction was indicated in the pictures by a clear, sharp interface between the salt and metal pool, and an increase in the pool depth. The salts were washed out of the cup with warm-running water. After oven drying, the reaction vessel and alloy product were weighed and the recovered neodymium was determined by the differences in weights. Core samples of the neodymium-iron alloy product were analyzed for chemical composition.

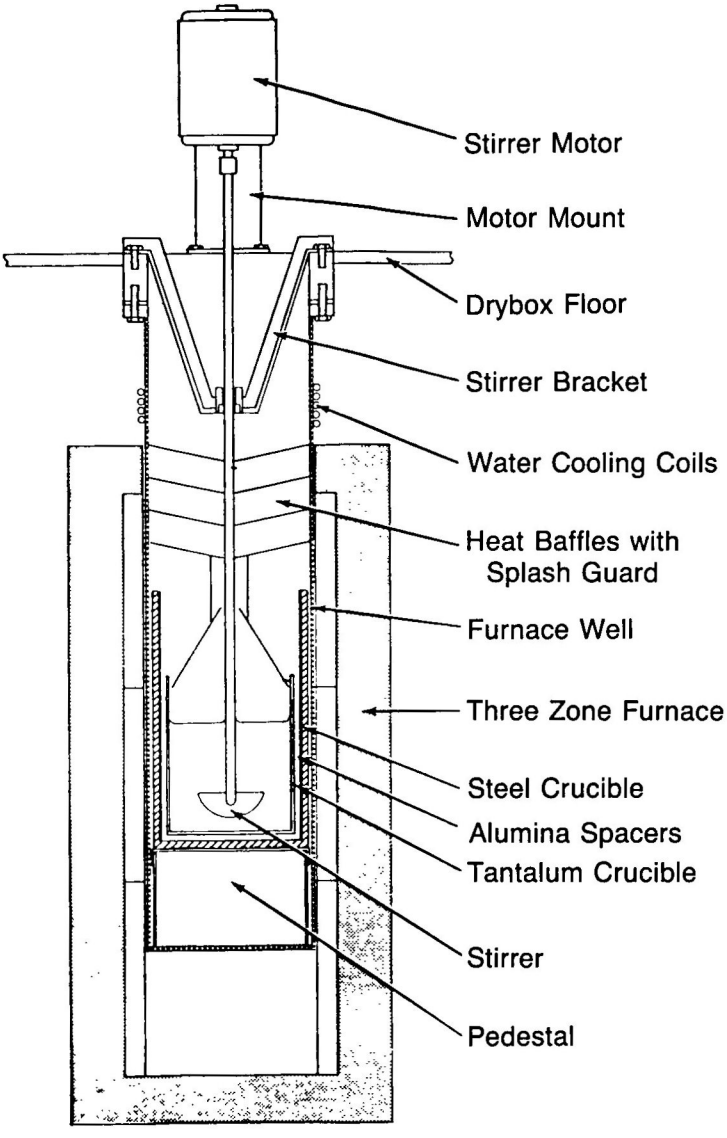


Fig. 2 Schematic diagram of the reduction apparatus.

Results and Discussion

The results of Nd_2O_3 reductions with sodium in $CaCl_2$-NaCl melts having 5 w/o KCl at 750°C are given in Table I. The mild steel reaction vessels had fins, and during mixing the stirrer blade was near the reactor bottom. The amount of Nd_2O_3 used per reduction run was 233.3 g (produces 200 g of Nd metal) except for Experiment 3 which used 221 g (189.5 g Nd) and Experiment 12 which used 221.6 g (190 g Nd). The salt ratio indicates the amount of salt per gram of neodymium that could theoretically be produced. The excess reductant is the percent of sodium above the stoichiometric amount which is 95.6 g to produce 200 g of neodymium. Recovery of neodymium is given as percent of the theoretical content of neodymium in the Nd_2O_3. Other variables are described in the table.

Clear separation between the salt phase and alloy pool was observed from the radiographs of the solidified products. A typical radiograph exhibiting good separation (having a sharp demarcation between the salt and metal phases) is shown in Fig. 3. The clear separation of the alloy from the salt phase was also indicated from the smooth surface of the solidified alloy after the salt was washed away. Examination by an optical microscope and by a scanning electron microscope of the samples taken from the solidified metallic pools did not indicate any salt inclusions (Fig. 4). Chemical analyses of the solidified metallic pools are given in Table II. The chemical analyses show the high quality of the Nd-Fe alloys produced. They are as good as those produced by using calcium reductant [3,4]. The yields were also as good as those achieved with the use of calcium reductant. A typical alloy pool from a successful reduction (Experiment 10; pool yield = 95.6%) contained 86.1 w/o Nd, 2.8 w/o Pr, 11.9 w/o Fe, less than 0.4 w/o Ca and very little other impurities.

General Observations

A systematic study was not conducted to determine the effects of the various parameters on the yield. But some parameters were judiciously manipulated to obtain a high yield (>90%) in the metallic pool. Data presented in Table I (Experiments 14-16, 22 and 23) indicate that 15% sodium in excess of the stoichiometric amount is sufficient to give a yield of 94.5% in the metallic pool under the experimental conditions reported. The results of Experiments 4-6, 10, 11 and 22 indicate that the ratio of salt/neodymium produced has to be about 4.5 to obtain a 95% yield in the metallic pool. The presence of KCl in the salt phase does not appear to influence the yield (Experiments 2 and 22). The data presented in Table III indicate that about 95% yield of neodymium in the metallic pool can be obtained using 900 g of salt with a 1.61 ratio of salt to reductant under the experimental conditions described (Experiment 15).

1	1200	6.0	10	503.9	84.0	81.4	9.3
2	1200	6.0	20	503.2	84.0	92.6	0.2
3	950	6.3	20	453.7	83.5	92.0	3.1
4	842	4.2	20	306.0	81.0	86.5	5.3
5	947	4.7	20	314.3	81.2	88.5	3.9
6	790	4.0	20	301.2	81.0	83.5	7.2
7	842	4.2	20	303.4	81.0	85.1	7.6
8	842	4.2	20	305.1	81.0	77.6	12.6
9	842	4.2	20	300.6	81.0	86.7	5.4
10	1000	5.0	20	301.4	82.1	95.6	0.1
11	1100	5.5	20	300.4	80.9	95.2	0.5
12	1000	5.3	27	300.0	80.9	93.4	1.2
13	1000	5.0	20	300.0	80.9	94.9	1.1
14	900	4.5	8	300.1	81.3	70.9	19.7
15	900	4.5	15	300.1	81.3	94.5	0.3
16	900	4.5	13	300.1	81.3	74.3	11.4
17	900	4.5	15	300.6	81.3	11.7	58.6
18	900	4.5	17	300.7	81.3	85.5	7.6
19	1000	5.0	20	301.8	81.3	89.2	4.1
20	1000	5.0	20	300.2	81.3	92.9	0.4
21	1100	5.5	20	300.6	81.3	93.8	0.6
22	900	4.5	20	300.5	80.9	92.3	0.5
23	900	4.5	17	300.5	80.9	92.9	0.3
24	900	4.5	17	301.4	81.0	90.3	0.5

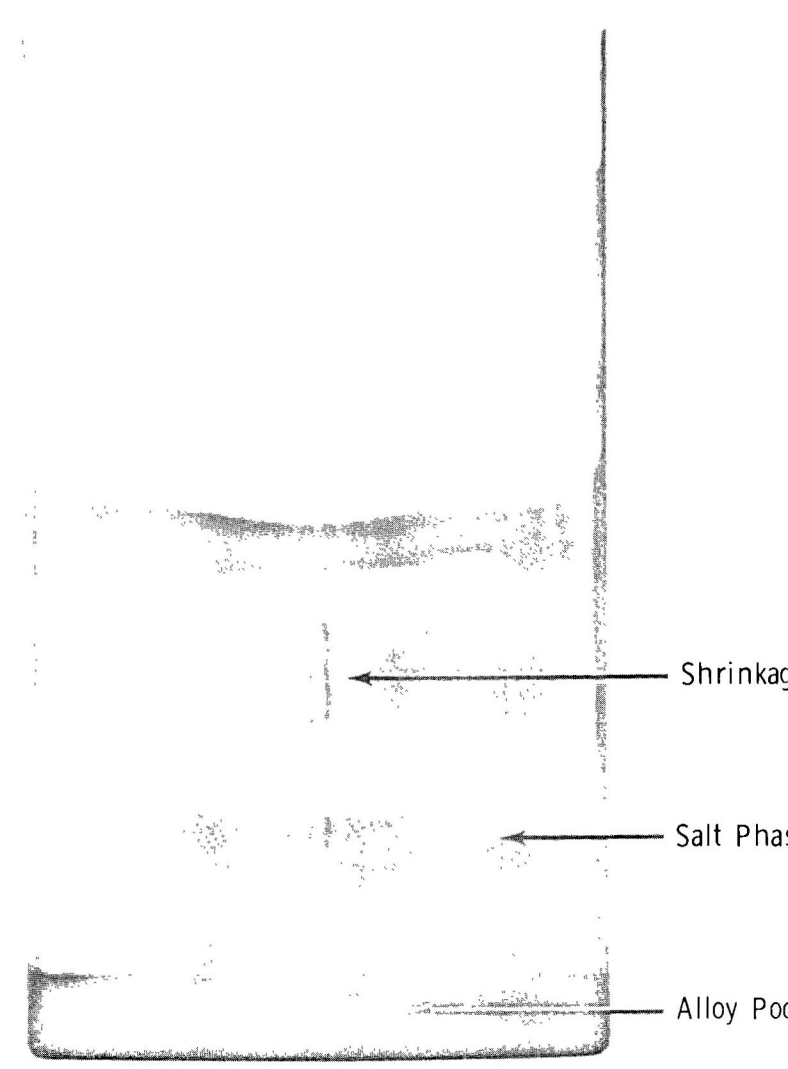
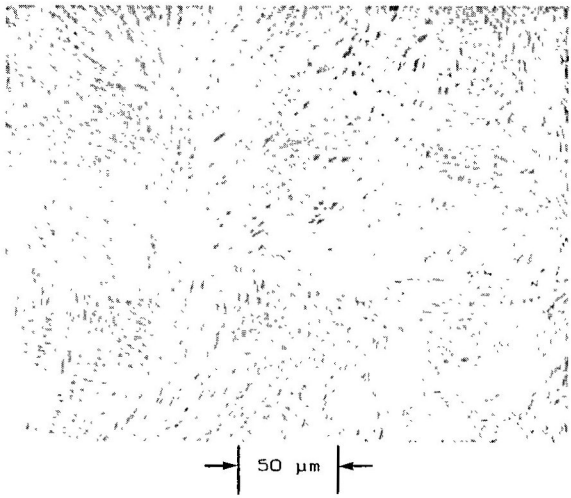


Fig. 3 Radiograph of reaction vessel from Experiment 10.



g. 4 Photomicrograph of a cross-section of the Nd-Fe pool from Experiment 10.

Table II. Chemical Analyses of Nd-Fe Alloy Samples Obtained from Sodium Reductions

iment	Alloy Analysis w/o Nd	w/o Fe	w/o Pr	Impurities (w/o) Al	Ca	Mg	Si
1	87.7	10.3		0.2	0.1	0.2	0.05
2	87.7	11.6	0.6	0.3	0.05	0.3	0.1
3	87.3	13.2	0.6	0.2	0.1	0.1	0.05
4	83.7	12.9		0.2	0.03	0.1	0.02
5	86.9	11.8	0.7	0.3	0.1	0.3	0.03
6	86.7	12.2		0.1	0.3	0.2	0.05
7	87.4	11.9		0.1	0.2	0.1	0.05
8	86.4	11.5	0.6	0.1	0.2	0.3	0.03
9	87.2	10.9	0.6	0.1	0.2	0.3	0.03
0	86.1	11.9	2.8	0.4	0.1	0.15	0.005
1	86.6	12.2		0.5	0.3	0.2	0.005
2	87.5	11.3		0.1	0.1	0.1	0.03
3	84.6	12.9		0.1	0.2	0.2	0.01
4	84.9	12.3		0.2	0.1	0.3	0.01
5	85.5	11.1		0.2	0.05	0.3	0.01
6	86.1	13.3		0.2	0.02	0.3	0.01
7	86.5	13.3	3.0	0.2	0.01	0.2	0.004
8	88.9	10.0	3.1	0.1	0.01	0.2	0.004
9	83.4	11.3		0.2	0.03	0.1	0.01
0	84.5	11.5		0.1	0.02	0.2	0.005
1	83.3	12.2	3.1	0.1	0.1	0.01	0.01
2	85.0	12.3		0.2	0.02	0.1	0.02
3	86.3	11.9		0.1	0.03	0.1	0.02
4	86.0	11.6		0.05	0.05	0.4	0.05

Table III. Effect of the $CaCl_2$/Na Ratio on Neodymium Recovery

Experiment	Total Salt Amount (g)	Salt-Reductant Ratio ($CaCl_2$/Na) (mole/atom)	Nd Recovery Total (g)	Nd Recovery In Pool (w/o)	Nd Recovery Nodules (w/o)
4	842	1.43	183.6	86.5	5.3
5	947	1.62	184.8	88.5	3.9
6	790	1.35	181.4	83.5	7.2
7	842	1.44	185.3	85.1	7.6
8	842	1.44	180.4	77.6	12.6
9	842	1.44	184.2	86.7	5.4
10	1000	1.72	191.4	95.6	0.1
11	1100	1.89	191.4	95.2	0.5
12	1000	1.72	179.7	93.4	1.2
13	1000	1.71	192.0	94.9	1.1
14	900	1.71	181.3	70.9	19.7
15	900	1.61	189.7	94.5	0.3
16	900	1.65	171.5	74.3	11.4
17	900	1.61	140.6	11.7	58.6
18	900	1.58	186.3	85.5	7.6
19	1000	1.72	186.6	89.2	4.1
20	1000	1.72	186.7	92.9	0.4
21	1100	1.89	188.8	93.8	0.6
22	900	1.54	185.6	92.3	0.5
23	900	1.58	186.4	92.9	0.3
24	900	1.58	181.5	90.3	0.5

According to the data presented in Ref. 2, two melts of Ca-Na coexist with a 70 w/o $CaCl_2$-30 w/o NaCl melt above 710°C under equilibrium conditions. Only one melt of Ca-Na will coexist with a $CaCl_2$-NaCl melt of $CaCl_2$ concentration higher than 70 w/o. In our experiments, pure sodium is added to a melt of 95 w/o $CaCl_2$-5 w/o KCl containing Nd_2O_3 in suspension. Under these conditions, a Na-Ca melt of higher sodium concentration will form first by reaction (3). The reaction will continue to form a Ca-Na melt of higher calcium concentration which will be in equilibrium with a $CaCl_2$-NaCl melt of $CaCl_2$ concentration greater than 70 w/o in the absence of Nd_2O_3. But with the formation of a Ca-Na melt of higher calcium concentration, the reduction reaction (2) will start to occur consuming the calcium generated by reaction (3). In this way, a $CaCl_2$-NaCl melt of 70 w/o $CaCl_2$ will form in the end and reaction (2) should proceed to completion to convert all the Nd_2O_3 into neodymium metal. Difficulty arises in having the proper amount of sodium metal. Sodium metal has to be taken in excess of the stoichiometric amount to compensate for the loss by vaporization. But if it is too much in excess, then near the end of reduction, a Na-Ca melt of higher sodium concentration will form. At this stage of reduction, a Ca-Na melt of higher calcium concentration is needed to scavenge all the NdOCl in the salt [3,4]. Therefore, the Na-Ca melt of higher sodium concentration may not be able to complete the reduction reaction. Because of this experimental difficulty, neodymium yields in Table I do not show any regular increase with the amount of sodium reductant.

On the other hand if you increase the $CaCl_2$ content of the salt phase while using the same amount of sodium, some of the calcium produced by reaction (3) is consumed through calcium solubility in the salt and thus enough calcium may not be available for the reduction. Attempts were made to obtain a high yield of neodymium in the metallic pool by varying the $CaCl_2$ and sodium ratio (Table III). However, much success was not achieved because of erratic loss of sodium. This type of problem will not be faced when the proper type of equipment is used on a larger scale.

Effect of Salt Ratio on Yield

One of the objectives of this study was to reduce the amount of salt per gram of neodymium produced (i.e., salt ratio) by changing the salt composition and/or decreasing the amount of excess sodium. The results of these reductions at 750°C (except Experiment 29 at 800°C) are given in Table IV. In Experiments 25-31 the reaction vessels had fins to assist in mixing; while in Experiments 32-37, they had no fins. As before, the amount of Nd_2O_3 per reduction run was 233.3 g which produces 200 g of neodymium metal, and therefore the salt ratio is the amount of salt divided by 200. The initial salt phase was 95 w/o $CaCl_2$-5 w/o KCl. The extraction pools consisted of ~300 g of 82.9 w/o Nd-Fe alloy. The stirrer blade was positioned ~0.1 cm above the bottom of the reactor except in Experiments 28 and 36, when the distance was ~2.5 cm, and Experiment 37 in which the distance was ~1.2 cm. In most cases, fast stirring was carried out at 600 rpm and the slow stirring at 120 rpm. However, in Experiments 32, 33 and 37, the fast stirring was done at 450 rpm; and in Experiment 35, it was carried out at 350 rpm.

Table IV. Description of the Experiments of Nd_2O_3 Reduction with Na-Metal for Determining the Effect of Salt Amount on the Yield

Experiment	Salt Amount (g)	Excess Reductant (w/o)	Stirring Times Fast (m)	Stirring Times Total (m)	Nd Recovery In Pool (w/o)	Nd Recovery Nodules (w/o)
25	750	15	55	70	94.8	0.4
26	700	10	55	70	90.4	2.1
27	650	10	55	70	79.8	3.8
28	700	10	55	70	90.9	0.4
29	700	10	55	70	92.8	0.8
30	650	10	55	70	93.2	1.6
31	700	10	105	120	2.2	80.2
32	750	15	55	70	90.1	1.7
33	750	15	105	120	93.6	0.6
34	750	15	55	70	89.1	3.8
35	750	15	105	120	90.4	1.3
36	750	15	55	70	88.8	1.0
37	750	15	105	120	84.4	8.0

Total neodymium yields in most of these experiments of greater than 90% and in some cases reaching ~95% were achieved. In some of these experiments neodymium yields in the Nd-Fe pool were also approaching 94%. This means extraction of the neodymium product from the salt phase was also good. These yields are comparable with those obtained with calcium reductant [3,4].

Chemical analyses of the solidified metallic pools are given in Table V. The chemical analyses show the high quality of the Nd-Fe alloys produced. A typical alloy contained 84.9 w/o Nd, 1.5 w/o Pr, 12.2 w/o Fe, <0.1 w/o Ca and very little other impurities when the yield in the pool was ~95% (Experiment 25).

Table V. Chemical Analyses of Nd-Fe Alloy Samples Obtained from Sodium Reductions

	Alloy Analysis			Impurities (w/o)			
Experiment	w/o Nd	w/o Fe	w/o Pr	Al	Ca	Mg	Si
25	84.9	12.2	1.5	0.2	0.05	0.1	0.005
26	86.1	12.5	1.5	0.1	0.05	0.1	0.005
27	84.9	12.6	1.6	0.2	0.1	0.2	0.005
28	85.4	12.8	1.7	0.1	0.05	0.1	0.005
29	83.9	12.8	1.6	0.1	0.1	0.1	0.003
30	84.4	12.5	1.6	0.1	--	0.05	0.003
31	--	--	--	--	--	--	--
32	85.2	12.8	0.9	0.05	0.05	0.05	0.002
33	86.8	12.0	1.7	0.1	0.05	0.03	0.002
34	87.4	12.4	1.7	0.05	0.05	0.03	0.002
35	86.1	11.9	1.1	0.05	0.05	0.05	0.003
36	86.1	11.5	1.1	0.05	0.05	0.05	0.003
37	86.0	12.7	1.6	0.1	0.01	0.05	0.01

The data from Nd_2O_3 reductions with different salt ratios (Table IV, Experiments 25-27, 29 and 30) indicate that the yield in the Nd-Fe extraction pool, other conditions being similar, decreases with decrease in salt ratio. While ~95% yield was observed at a salt ratio of 3.75, only 80% was recovered in the pool at 3.25. Included in these experiments are the effects of temperature and mixed reductant on neodymium yield. When the temperature was raised from 750 to 800°C (Experiment 29), at a salt ratio of 3.5, the pool yield rose from 90 to ~93%. In addition, the salt ratio was successfully reduced to 3.25 (~95% total yield) when calcium (11.5 w/o) was added to the sodium reductant (Experiment 30).

Effect of Stirrer Position and Fins on Yield

As shown in Table IV, the paddle position in the reactor does not appear to influence the yield of neodymium in the pool. The yield in Experiment 28 when the paddle was positioned well up in the melt (2.5 cm above the vessel

bottom) was essentially the same as that for Experiment 26 when the paddle was placed directly in the extraction pool (0.1 cm above the vessel bottom). However, under similar experimental conditions, the yield appears to decrease when fins are not used in the reactor (Table IV, 25 and 34 or 28 and 36). But by increasing the reaction time and adjusting the stirring rate, a higher yield can be obtained in a reactor without fins (Table IV, 33). In the case of reactors with fins, it appears that an increase in reaction time caused the yield to decrease (Table IV, 26 and 31). Evidently, more sodium evaporates causing incomplete reduction of the NdOCl which is necessary for good separation [5,6]. When more sodium was available, longer reactions times improved yield (Table IV, 32 and 33). These observations indicate that a further systematic study is needed to develop a relationship between fins, stirring rate and stirrer shape to give a stirring pattern for obtaining the highest neodymium yields.

Acknowledgments

The authors wish to express their appreciation to Bud Verrier for assistance with the experimental work, to Noel Potter and his associates in the Analytical Chemistry Department for analyses of the alloy samples, and to Drs. Kathleen C. Taylor and John S. Dunning for helpful suggestions in preparing this manuscript.

References

1. J. J. Croat, J. F. Herbst, R. W. Lee and F. E. Pinkerton, J. Appl. Phys. 55(8), 2078 (1984).

2. R. A. Sharma, "Neodymium Production Processes," J. Metals. 39, 33, (1987).

3. R. A. Sharma and R. N. Seefurth, "Metallothermic Reduction of Nd_2O_3 with Ca in $CaCl_2$-KCl Melts," Joint International Symp. on Molten Salts, Oct. 18-23, 1987, Honolulu, HI.

4. R. A. Sharma and R. N. Seefurth, "Metallothermic Reduction of Nd_2O_3 with Ca in $CaCl_2$-NaCl Melts," J. Electrochem. Soc., 135, 66 (1988).

5. R. A. Sharma and R. N. Seefurth, "A Molten Salt Process for Producing Neodymium Metal," Annual AIME Meeting, Phoenix, AZ (1988).

6. R. A. Sharma and R. N. Seefurth, "A Molten Salt Process for Producing Neodymium Metal," Light Metals 1988, L. G. Boxall, editor, The Metallurgical Soc. Inc., Warrendale, PA (1987).

7. JANAF Thermochemical Tables, 2nd Edition, NSRDS-NBS, June 1971-1978.

8. I. Barin, O. Knacke, and O. Kubachewski, "Thermochemical Properties of Inorganic Substances," Supplement, Springer-Verlag, Berlin, Heidelberg, NY (1977).

9. W. G. Moffatt, "The Handbook of Binary Phase Diagrams," General Electric Company, 120 Erie Boulevard, Schenectady, NY 12305.